DIRECT DIGITAL MANUFACTURING

3D Printed Chopstick Holder

STL Download → SolidWorks Modification → FDM Print

Material: PLA (White) | Infill: 20% Grid | Speed: 40%

Project Objective To design, modify, and 3D print a functional chopstick holder that improves ease of use through an ergonomic finger-slot gripping feature. The original STL was downloaded from an online repository and found to be ill-fitting; SolidWorks was used to reverse-engineer and modify the geometry before re-printing.

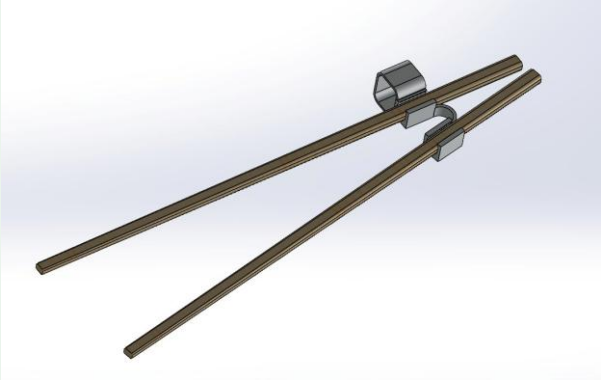
1. CAD vs Printed Model — Visual Comparison

FIGURE 1 — Original Downloaded STL



Standard chopstick pair — no holding feature

FIGURE 2 — Modified CAD Model (SolidWorks)



Re-engineered with finger-slot holding bracket





2. Design Modification Summary

Original STL (Downloaded)	Modified CAD (SolidWorks)	Purpose of Change
Two plain square-section chopstick rods	Same chopstick geometry retained	Preserve core form factor
No connector or joining feature	U-shaped bracket/clip added at pivot point	Allow two sticks to be held together
Fixed at a single junction — rigid	Hinged slot region allows slight spring flex	Enable open/close action for gripping
No ergonomic finger feature	Finger-slot cutout on each bracket half	Index finger inserts to control chopstick angle
STL not parametric — cannot be edited	SolidWorks sketch trace converts to editable solid	Enables dimension modification & iteration





3D PRINTED CHOPSTICK ITERTION ONE AND TWO REAL IMAGES SHOWN IN FIG

3. Design & Fabrication Workflow

	Stage	Description
01	**STL Sourcing**	Downloaded chopstick STL file from online repository (Thingiverse/Printables)
02	**Initial Slicing & Print**	Imported into Anycubic Slicer — PLA White, 20% Grid infill, 40% speed. Printed and tested fit.
03	**Problem Identified**	Printed chopsticks did not fit hand correctly; no holding mechanism to guide finger position
04	**STL → SolidWorks Conversion**	Imported STL into SolidWorks; used sketch trace on cross-section profile to rebuild as parametric solid body
05	**CAD Modification**	Designed finger-slot bracket: U-shaped clip body, curved inner channel to grip each stick, slot feature for index finger insertion
06	**Re-Print Modified Design**	Exported modified SolidWorks model as new STL; re-sliced and printed with same parameters

4. Print Parameters

Parameter	Value / Setting
Printer	Anycubic FDM 3D Printer
Material	PLA — White
Infill Density	20%
Infill Pattern	Grid
Print Speed	40% of max (slow — high accuracy)
Layer Height	0.2 mm (Standard)
Supports	Not required
Bed Temp	60 °C (PLA standard)
Nozzle Temp	200–215 °C
Software (Slicing)	Anycubic Slicer
CAD Software	SolidWorks (STL trace + modification)

5. Key Design Features of Modified Model

Feature	Design Intent
U-shaped Clip Bracket	Clamps around each chopstick rod; holds both sticks at correct pivot distance
Finger-Slot Cutout	Index finger inserts into slot to naturally control the open/close angle of chopsticks
Curved Inner Channel	Matches chopstick cross-section profile traced from STL for a snug, friction fit
Spring-flex Gap	Small gap between bracket halves allows slight flex so chopsticks can be pressed together and released
Parametric SolidWorks Body	STL profile traced to sketch → extruded solid → fully editable for future size adjustments

6. Challenges & Solutions

Challenge	Root Cause	Solution Applied
Original STL did not fit hand	Online model designed for generic dimensions, no ergonomic feature	Converted STL to SolidWorks solid and added custom finger-slot bracket
STL file not directly editable	STL is mesh data — no sketch or feature tree	Used SolidWorks STL import + sketch trace method to rebuild parametric profile
Bracket did not grip chopstick initially	Inner channel slightly oversized after tracing	Reduced channel dimension by 0.2 mm tolerance compensation; reprinted

✓

Conclusion

This project successfully demonstrated the full DDM workflow: **download → evaluate → redesign → re-manufacture**. The original STL chopstick model was non-functional for the user's hand dimensions. By importing the STL into SolidWorks, tracing the cross-section sketch, and building a parametric model, a custom **finger-slot holding bracket** was added. The modified design was sliced and printed successfully in PLA using the Anycubic slicer at low speed (40%) to maximise dimensional accuracy.

Key Outcomes: (1) Chopsticks now fit the user's hand. (2) Finger-slot guides correct grip. (3) PLA at 40% speed produced clean, well-defined bracket geometry. (4) SolidWorks STL-to-sketch workflow is viable for adapting existing 3D models.